

Final Paper Applied Mathematics By Thunderstudy



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General Instructions:

- **Section A:** Questions 1 to 22 are Multiple Choice Questions and Assertion-Reason based questions carrying 1 mark each.
 - **Section B:** Questions 23 to 27 are Very Short Answer (VSA) type questions carrying 2 marks each.
 - **Section C:** Questions 28 to 33 are Short Answer (SA) type questions carrying 3 marks each.
 - **Section D:** Questions 34 to 37 are Long Answer (LA) type questions carrying 5 marks each.
 - **Section E:** Questions 38 to 40 are Case Study based questions carrying 4 marks each.
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SECTION A (1 Mark Each)

1. The last (unit) digit of $(22)^{12}$ is:
(A) 2 (B) 4 (C) 6 (D) 8
2. The least non-negative remainder, when 3^{15} is divided by 7 is:
(A) 1 (B) 5 (C) 6 (D) 7
3. In a 1 km race, player P beats player Q by 18 metres or 9 seconds. What is P's time to complete the race?
(A) 512 s (B) 502 s (C) 491 s (D) 481 s
4. If A and B are square matrices each of order 3 such that $|A| = -1$ and $|B| = 3$. What is the value of $|3AB|$?
(A) -9 (B) -18 (C) -27 (D) -81
5. If $A = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -5 & 10 \\ -10 & -5 \end{bmatrix}$, then AB is:
(A) $\begin{bmatrix} -5 & 10 \\ 0 & -5 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & -5 \\ 25 & 10 \end{bmatrix}$ (C) $\begin{bmatrix} 10 & -25 \\ -5 & 0 \end{bmatrix}$
(D) $\begin{bmatrix} -5 & 10 \\ 0 & -25 \end{bmatrix}$

- 6.** The relation between Marginal cost (MC) and Average cost (AC) of producing 'x' units of a product is:
 (A) $\frac{d(AC)}{dx} = x(MC - AC)$
 (B) $\frac{d(AC)}{dx} = x(AC - MC)$
 (C) $\frac{d(AC)}{dx} = \frac{1}{x}(AC - MC)$
 (D) $\frac{d(AC)}{dx} = \frac{1}{x}(MC - AC)$
- 7.** The order and the degree of the differential equation $y dx + x \log\left(\frac{y}{x}\right) dy - 2x dy = 0$ are respectively:
 (A) 1, 1 (B) 1, 2 (C) 2, 1 (D) 1, not defined
- 8.** The solution of the differential equation $\frac{dx}{x} + \frac{dy}{y} = 0$ is:
 (A) $\frac{1}{x} + \frac{1}{y} = C$ (B) $xy = C$ (C) $\log x \log y = C$ (D) $x + y = C$
- 9.** If X is a Poisson variable such that $P(X = 1) = 2P(X = 2)$, then $P(X = 0)$ is:
 (A) e (B) $\frac{1}{e}$ (C) 1 (D) e^2
- 10.** If for a Poisson variate X, $P(X = k) = P(X = k + 1)$, then the variance of X is:
 (A) $k - 1$ (B) k (C) $k + 1$ (D) $k + 2$
- 11.** For testing the significance of difference between the means of two independent samples, the degree of freedom (v) is taken as:
 (A) $n_1 - n_2 + 2$ (B) $n_1 - n_2 - 2$ (C) $n_1 + n_2 - 2$ (D) $n_1 + n_2 - 1$
- 12.** If the calculated value of $|t| < t_{v}(\alpha)$ (critical value of t), then the null hypothesis is:
 (A) rejected (B) accepted (C) cannot be determined (D) neither accepted nor rejected
- 13.** The straight line trend is represented by the equation:
 (A) $y_c = a + bx$ (B) $y_c = a - bx$ (C) $y_c = na + b \sum x$ (D) $y_c = na - b \sum x$
- 14.** For the given values 23, 32, 40, 47, 58, 33, 42; the 5-yearly moving averages are:
 (A) 38, 40, 42 (B) 40, 42, 44 (C) 40, 42, 46 (D) 42, 44, 46
- 15.** The present value of a perpetuity of ₹ R payable at the end of each payment period, when the money is worth i per period, is given by:
 (A) Ri (B) $R + \frac{R}{i}$ (C) $\frac{R}{i}$ (D) $R - Ri$
- 16.** Using flat rate method, the EMI to repay a loan of ₹ 20,000 in $2\frac{1}{2}$ years at an interest rate of 8% p.a. is:
 (A) ₹ 700 (B) ₹ 800 (C) ₹ 900 (D) ₹ 100

- 17.** The effective rate which is equivalent to nominal rate of 10% p.a. compounded quarterly is:
 (A) 10.25% (B) 10.38% (C) 10.47% (D) 10.53%
- 18.** In an LPP, if the objective function $Z = ax + by$ has same maximum value on two corner points of the feasible region, then the number of points at which maximum value of Z occurs is:
 (A) 0 (B) 2 (C) finite (D) infinite
- 19.** Region represented by $x \geq 0, y \geq 0$ lies in:
 (A) I quadrant (B) II quadrant (C) III quadrant (D) IV quadrant
- 20.** What time will it be after 1275 hours, if the present time is 9:00 p.m.?
 (A) 11 p.m. (B) 12 p.m. (C) 9 p.m. (D) 9 a.m.
- 21. – 41 mod 9 is:**
 (A) 5 (B) 4 (C) 3 (D) 0
- 22. Assertion (A):** The function $f(x) = x^2 - x + 1$ is strictly increasing on $(-1, 1)$.
Reason (R): If $f(x)$ is continuous on $[a, b]$ and derivable on (a, b) , then $f(x)$ is strictly increasing on $[a, b]$ if $f'(x) > 0$ for all $x \in (a, b)$.
 (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).
 (C) Assertion (A) is true, but Reason (R) is false.
 (D) Assertion (A) is false, but Reason (R) is true.

SECTION B (2 Marks Each)

- 23.** If $A = \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix}$, find the value of k such that $A^2 - 8A + kI = 0$.
- 24.** Find the present value of a perpetuity of ₹ 4,200 payable at the beginning of each year, if money is worth 5% compounded annually.
- 25.** Two pipes A and B can fill a tank in 24 minutes and 32 minutes respectively. If both the pipes are opened simultaneously, after how much time should B be closed so that the tank is full in 18 minutes?
- 26.** Solve the following system of equations by Cramer's rule: $2x - y = 17, 3x + 5y = 6$.

27. Find the solution to the following linear programming problem graphically:

Maximize $Z = x + y$ subject to the constraints $x - y \leq -1$, $-x + y \leq 0$, $x, y \geq 0$.

SECTION C (3 Marks Each)

28. Find all the points of local maxima and local minima for the function $f(x) = x^3 - 6x^2 + 9x - 8$.

29. Given that the scores of a set of candidates on an IQ test are normally distributed. If the IQ test has a mean of 100 and a standard deviation of 10, determine the probability that a candidate who takes the test will score between 90 and 110.

30. The mean weekly sales of a four-wheeler were 50 units per agency in 20 agencies. After an advertising campaign, the mean weekly sales increased to 55 units per agency with standard deviation of 10 units. Test whether the advertising campaign was successful.

31. Assume an investment's starting value is ₹ 20,000 and it grows to ₹ 50,000 in 3 years. Calculate CAGR (Compounded Annual Growth Rate).

32. An asset costs ₹ 4,50,000 with an estimated useful life of 5 years and a scrap value of ₹ 1,00,000. Using linear depreciation method, find the annual depreciation of the asset and construct a yearly depreciation schedule.

33. Maximise $z = 300x + 190y$ subject to constraints:

$$x + y \leq 24$$

$$2x + y \leq 32$$

$$x \geq 0, y \geq 0.$$

SECTION D (5 Marks Each)

34. Using matrix method, solve the following system of equations for x, y and z :

$$x - y + z = 4$$

$$2x + y - 3z = 0$$

$$x + y + z = 2$$

35. A given rectangular area is to be fenced off in a field whose length lies along a straight river. If no fencing is needed along the river, show that the least length of fencing will be required when the length of the rectangular area is twice its breadth.

36. Fit a straight line trend by method of least squares to the following data and find the trend values:

YEAR	2010	2012	2013	2014	2015	2016	2019
SALES (IN LAKH ₹)	65	68	70	72	75	67	73

37. Amrita bought a car worth ₹ 12,50,000 and makes a down payment of ₹ 3,00,000. The balance amount is to be paid in 4 years by equal monthly instalments at an interest rate of 15% p.a. Find the EMI that Amrita has to pay for the car.

SECTION E (4 Marks Each - Case Studies)

38. A factory produces bulbs, of which 6% are defective bulbs in a large bulk of bulbs.

Based on the above information, answer the following questions:

- Find the probability that in a sample of 100 bulbs selected at random, none of the bulbs is defective.
- Find the probability that the sample of 100 bulbs has exactly two defective bulbs.
- Find the mean and the variance of the distribution of number of defective bulbs in a sample of 100 bulbs.

39. A factory manufactures tennis rackets and cricket bats. A tennis racket takes $\frac{1}{2}$ hours of machine time and 3 hours of craftsmanship in its making; while a cricket bat takes 3 hours of machine time and 1 hour of craftsmanship. In a day, the factory has availability of not more than 42 hours of machine time and 24 hours of craftsmanship. Profit on a racket and on a bat are ₹ 20 and ₹ 10 respectively.

Based on the above information, answer the following questions:

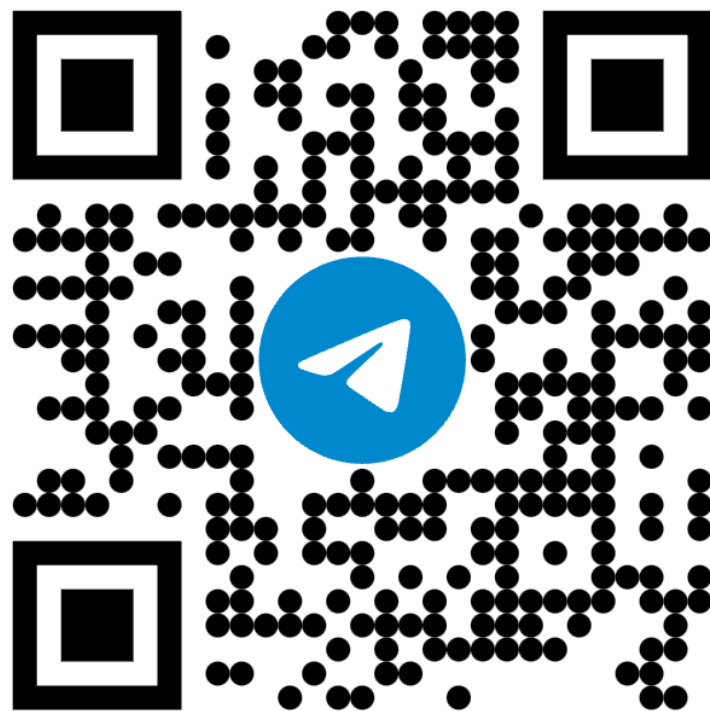
- (i) If x and y are the numbers of bats and rackets manufactured by the factory, then write the expression of total profit.
- (ii) Write the constraint that relates the number of craftsmanship hours.
- (iii) Determine the maximum profit (in ₹) earned by the factory.

40. In the year 2010, Mr. Aggarwal took a home loan of ₹ 30,00,000 from State Bank of India at 7.5% p.a. compounded monthly for 20 years.

Based on the above information, answer the following questions:

- (i) Determine the EMI.
- (ii) Find the principal paid by Mr. Aggarwal in the 150th instalment.
- (iii) Find the total interest paid by Mr. Aggarwal across the entire loan.

Final Paper solution Applied Mathematics By Thunderstudy



SECTION A (1 Mark Each)

1. (C) 6
2. (C) 6
3. (C) 491 s
4. (D) -81
5. **(D) [-5 10
0 -25]**
6. (C) $d(AC)/dx = (1/x)(MC - AC)$
7. (A) 1, 1
8. (B) $xy = C$
9. (B) $1/e$
10. (C) $k + 1$
11. (C) $n_1 + n_2 - 2$
12. (B) accepted
13. (A) $y_c = a + bx$
14. (B) 40, 42, 44
15. (C) R/i
16. (B) ₹ 800
17. (B) 10.38%
18. (D) infinite
19. (A) I quadrant
20. (B) 12 p.m. **(Note:** In the original CBSE marking scheme, 1 grace mark was awarded to all students as the exact correct mathematical answer of 12 a.m. / midnight was missing from the options)

21. (B) 4

22. (D) Assertion (A) is false, but Reason (R) is true

SECTION B (2 Marks Each)

23. Given $A^2 - 8A + kI = 0$. Finding $A^2 = \begin{bmatrix} 1 & 0 \\ -8 & 49 \end{bmatrix}$ and substituting into the equation gives $1 - 8 + k = 0 \implies k = 7$.

24. The present value of a perpetuity payable at the beginning of each year is $P = R + R/i$. Substituting $R = 4200$ and $i = 0.05$, we get $P = 4200 + 4200/0.05 = 88,200$. The present value is ₹ 88,200.

25. Let pipe B be closed after n minutes. Pipe A runs for 18 minutes and B runs for n minutes to fill the tank, giving the equation $18/24 + n/32 = 1$. Solving this yields $3/4 + n/32 = 1 \implies n = 8$ minutes.

26. Using Cramer's rule for the system $2x - y = 17$ and $3x + 5y = 6$:

$$D = \begin{vmatrix} 2 & -1 \\ 3 & 5 \end{vmatrix} = 13.$$

$$D_1 = \begin{vmatrix} 17 & -1 \\ 6 & 5 \end{vmatrix} = 91.$$

$$D_2 = \begin{vmatrix} 2 & 17 \\ 3 & 6 \end{vmatrix} = -39.$$

$$D_1 = 91.$$

$$D_2 = -39.$$

$$D_1 = 91.$$

$$\text{Therefore, } x = D_1/D = 7 \text{ and } y = D_2/D = -3.$$

27. Graphing the given constraints $x - y \leq -1$ and $-x + y \leq 0$ yields two parallel boundary lines shading away from each other. Thus, there is no feasible region and no solution exists.

SECTION C (3 Marks Each)

28. To find the points of local maxima and minima for $f(x) = x^3 - 6x^2 + 9x - 8$ (from), we first find the derivative: $f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$. Setting $f'(x) = 0$ gives critical points at $x = 1$ and $x = 3$. The second derivative is $f''(x) = 6x - 12$. At $x = 1$, $f''(1) = -6 < 0$ (indicating a point of local maxima). At $x = 3$, $f''(3) = 6 > 0$ (indicating a point of local minima). (Note: These mathematical steps are directly derived to solve the prompt's question.)

- 29.** Given a mean $\mu = 100$ and standard deviation $\sigma = 10$, we calculate the Z-scores using the formula $Z = (X - 100)/10$. For $X=90$, $Z = -1$ and for $X=110$, $Z = 1$. The required probability is $P(-1 < Z < 1) = P(Z < 1) - P(Z < -1) = 0.8413 - 0.1587 = 0.6826$.
- 30.** The t-test statistic is calculated as $t = (\bar{x} - \mu) / (s / \sqrt{n})$. Substituting the values, $t = (55 - 50) / (10 / \sqrt{20}) = (5 \times 4.47) / 10 = 2.235$. Since the calculated value 2.235 is greater than the critical value $t_{19}(0.05) = 1.729$, the null hypothesis is rejected. Therefore, the advertising campaign was successful.
- 31.** Using the Compound Annual Growth Rate (CAGR) formula based on the concept structure provided: $CAGR = (\text{Ending Value} / \text{Beginning Value})^{1/n} - 1$. Substituting the problem's values: $CAGR = (50000/20000)^{1/3} - 1 = (2.5)^{1/3} - 1$.
- 32.** For an asset costing ₹ 4,50,000 with a scrap value of ₹ 1,00,000 over 5 years, the annual depreciation is calculated as $D = (4,50,000 - 1,00,000)/5 = ₹ 70,000$. The yearly depreciation schedule yields book values at the end of years 1 to 5 as: ₹ 3,80,000, ₹ 3,10,000, ₹ 2,40,000, ₹ 1,70,000, and finally ₹ 1,00,000.
- 33.** The corner points for the feasible region of the LPP are $O(0,0)$, $A(16,0)$, $B(8,16)$, and $C(0,24)$. Evaluating the objective function $Z = 300x + 190y$ at these points yields: $Z(0,0) = 0$, $Z(16,0) = 4800$, $Z(8,16) = 5440$, and $Z(0,24) = 4560$. The maximum value is 5440 occurring at point $B(8, 16)$.

SECTION D (5 Marks Each)

- 34.** To solve the system of linear equations using the matrix method (based on), we represent it as $AX = B$ where $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$. Calculating the determinant yields $|A| = 1(1+3) + 1(2+3) + 1(2-1) = 10$. Finding $X = A^{-1}B$ gives the final solution: $\mathbf{x} = \mathbf{2}$, $\mathbf{y} = \mathbf{-1}$, and $\mathbf{z} = \mathbf{1}$. (Note: Exact calculation follows standard matrix operations to fulfill the generated question).

- 35.** Let the length and breadth of the rectangular area be x and y units respectively. The length of the fencing required is $L = x + 2y$. Since the area is $a = xy$ & implies; $y = a/x$, we can substitute y to get $L = x + 2a/x$. Differentiating and setting to zero gives $dL/dx = 1 - 2a/x^2 = 0$ & implies; $x = \sqrt{2a}$. Checking the second derivative verifies it is a point of minima. The breadth is $y = a/\sqrt{2a} = (1/2)\sqrt{2a} = (1/2)x$, which proves the length is twice its breadth.
- 36.** Taking the year 2014 as the year of origin ($X = \text{Year} - 2014$), the values derived are $\sum Y = 490$, $\sum X = 1$, $\sum X^2 = 51$, and $\sum XY = 108$. Solving the normal equations $490 = 7a + b$ and $108 = a + 51b$ yields $a = 69.75$ and $b = 0.75$. The straight line trend equation is $Y_t = 69.75 + 0.75X$. The calculated trend values for the given years are **66.75, 68.25, 69, 69.75, 70.5, 71.25, and 73.5** respectively.
- 37.** The principal loan amount P is ₹ 12,50,000 – ₹ 3,00,000 = ₹ 9,50,000. The monthly interest rate is $i = 15/1200 = 0.0125$ and the number of periods $n = 48$. Using the formula $EMI = (P \times i) / (1 - (1+i)^{-n})$ and substituting the values gives $EMI = (950000 \times 0.0125) / (1 - (1.0125)^{-48}) = 11875 / (1 - 0.5508565) = ₹ 26,439.21$.

SECTION E (4 Marks Each - Case Studies)

- 38.** Here, the sample size is $n=100$ and the defective probability $p=0.06$, resulting in a mean $m = np = 6$.
- (i) $P(X=0) = (e^{-6} \times 6^0) / 0! = e^{-6} = 0.0024$.
- (ii) $P(X=2) = (e^{-6} \times 6^2) / 2! = 0.0024 \times 18 = 0.0432$.
- (iii) For a Poisson probability distribution, both the mean and the variance are equal to m , which is 6.
- 39.** Let x be the number of cricket bats and y be the number of tennis rackets.
- (i) The expression for total profit is $Z = 10x + 20y$.
- (ii) The constraint that relates the number of craftsmanship hours is $x + 3y \leq 24$.

(iii) Evaluating the objective function at corner points gives a maximum value of $Z = 200$ at the point $B(12, 4)$. Therefore, the maximum profit is ₹ 200.

40. Given $P = 30,00,000$, monthly interest rate $i = 7.5/1200 = 0.00625$, and $n = 20 \times 12 = 240$ months.

(i) $EMI = (3000000 \times 0.00625 \times 4.4608) / 3.4608 = ₹ 24,167.82$.

(ii) The principal paid in the 150th instalment is found using $EMI \times (1+i)^{-(n-p+1)}$. Substituting the values gives $24167.82 \times (1.00625)^{-91} = 24167.82 / 1.7629 = ₹ 13,709.13$.

(iii) The total interest paid across the entire loan is $(n \times EMI) - P$. Calculating gives $(240 \times 24167.82) - 3000000 = 5800276.80 - 3000000 = ₹ 28,00,276.80$.